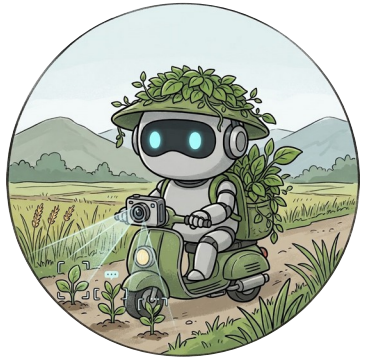


NCHC Open Hackathon

Final Presentation

CropSight

2026/11/04

Mentor



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Members



賴聖傑



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NVIDIA

TOOLMEN

機器學習與機器視覺實驗室

Lab of Machine Learning and Machine Vision

OpenACC More Science, Less Programming

OPEN HACKATHONS

NCHC

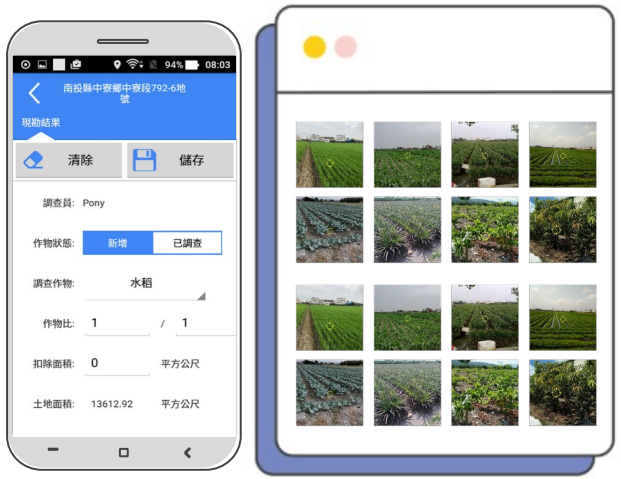
NARLabs 財團法人國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

NVIDIA

Problem

Current Survey Limitations

- Fragmented farmland.
- Diverse terrains.
- 300+ crop types.



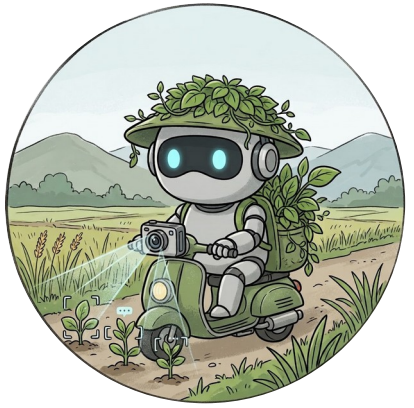
The Cost of Scaling

Past
Handheld Sampling



✓ High Precision
✗ Low Scalability
✗ Low Efficiency

Current
GoPro with motor



✓ High Efficiency
✓ High Scalability
✗ Low Precision

CropSight

Crop identification in street-view imagery

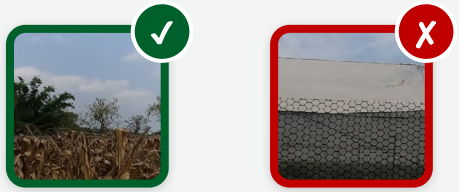
1 ICCA
Image Center Crop

Crop & resize to 384×384



2 NCIFM
Non-Crop Image Filtering

SegFormer-B4 (150 classes)
ExG greenness + CLIP
→ vegetation ratio calculation



3 TCCFM
Target Crop Classification

ViT-L/16 (21 crops)
→ target / uncertain / non-target



Why acceleration matters

million images
Manual survey took months to years

Algorithm

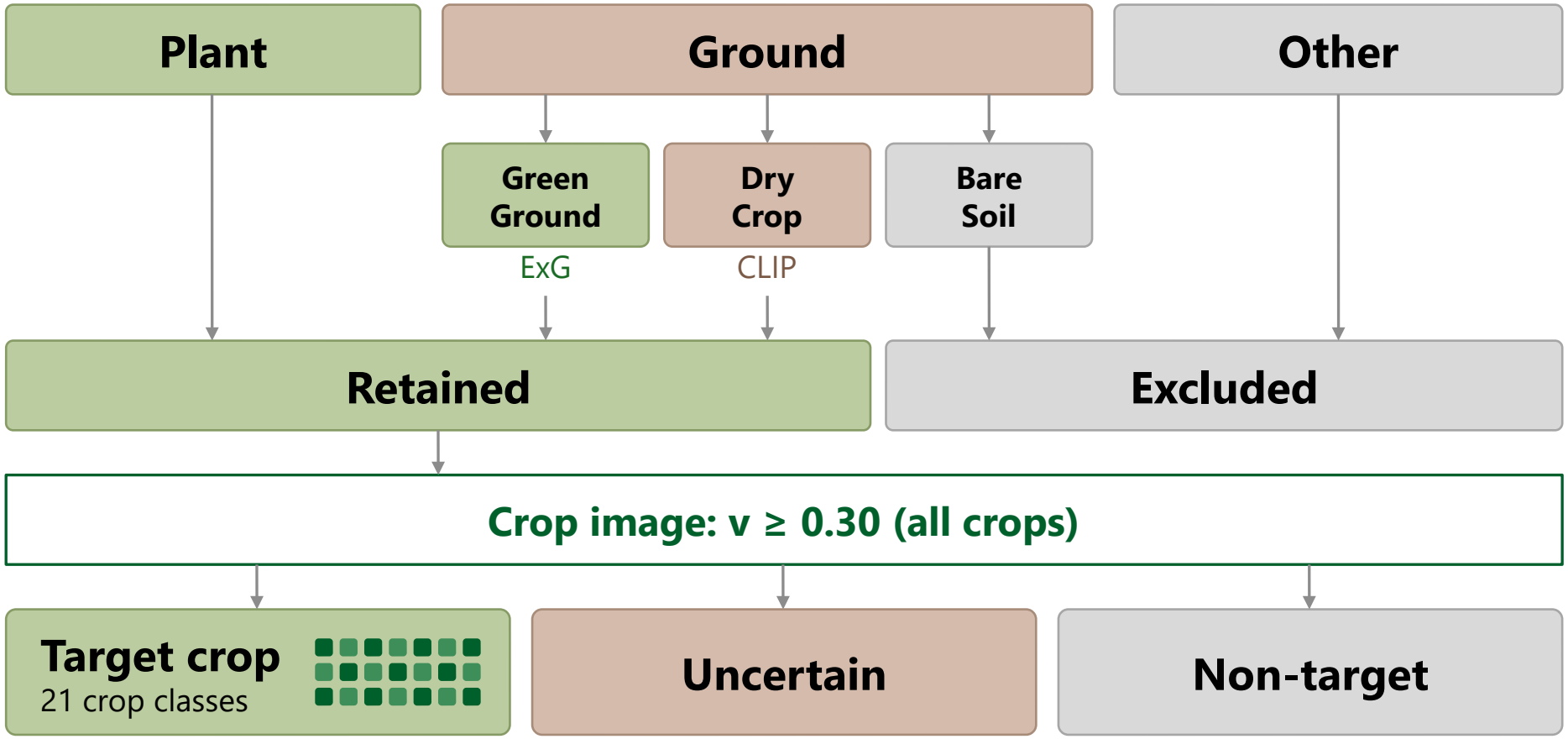
SegFormer · CLIP · ViT inference
PyTorch · TensorRT

What we focused on

Avg. time per batch, one GPU
Same results as baseline

Pipeline

1. **SegFormer-B4**
   Labels every pixel
2. **ExG + CLIP**
   Re-checks ground pixels
3. **Grouping**
   Vegetation vs. the rest
4. **Vegetation ratio**
   Decides crop image
5. **TCCFM · ViT-L/16**
   Classifies 21 crops



Evolution and Strategy

1 **Goal** — Faster pipeline on one GPU, same predictions

2 **Initial strategy** — CPU preprocessing took ~65% of time → move it to the GPU

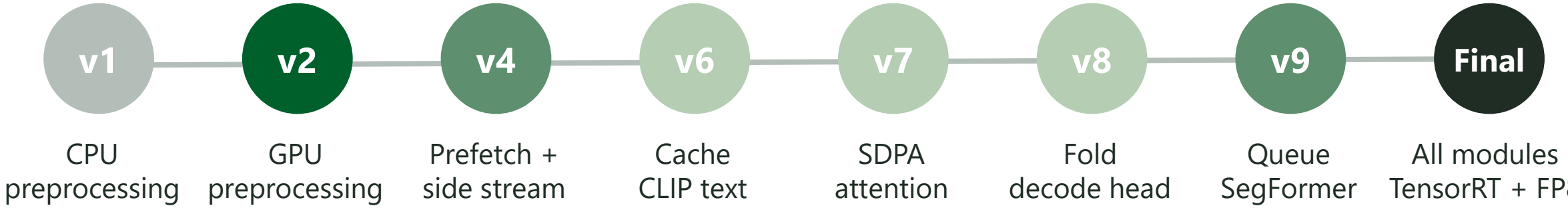
3 **How it changed** — Bottleneck moved: CPU prep → Image quick load→ GPU compute

Our loop for every version



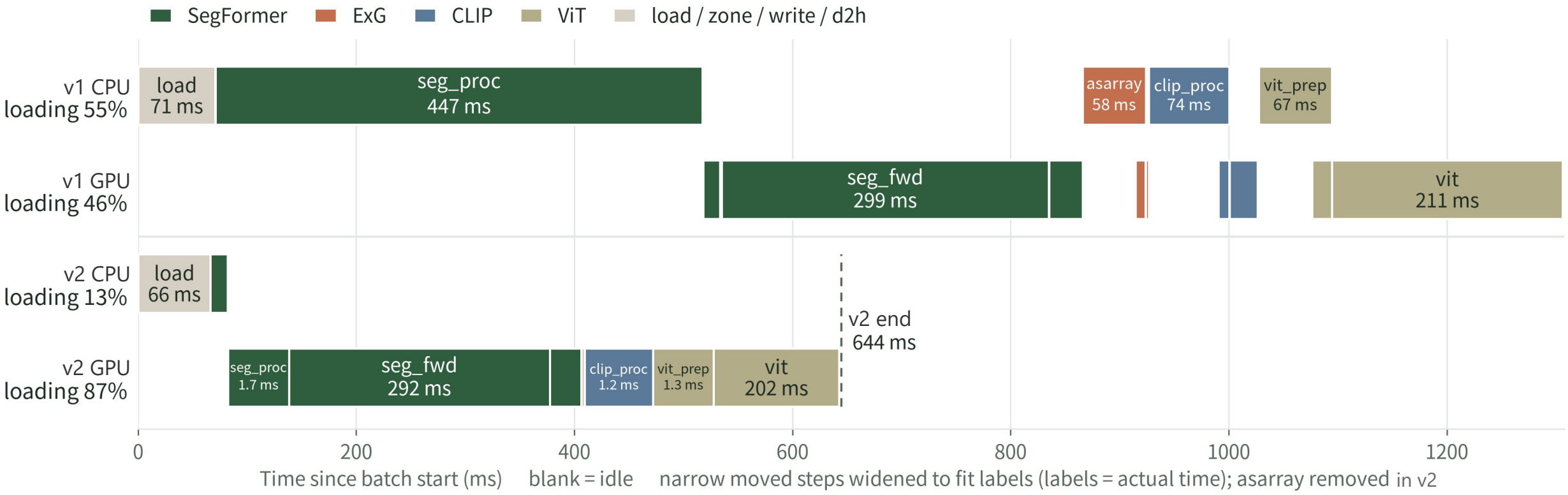
Our Journey: v1 → Final

Move work to GPU Overlap CPU & GPU Multi Process TensorRT + FP8



1306	156.1	139.4	145.6	137.9	130.2	115.8	76.5
Baseline	8.4×	9.4×	9.0×	9.5×	10.0×	11.3×	17.1×
TITAN RTX	H200						

Our Journey: v1 → v2

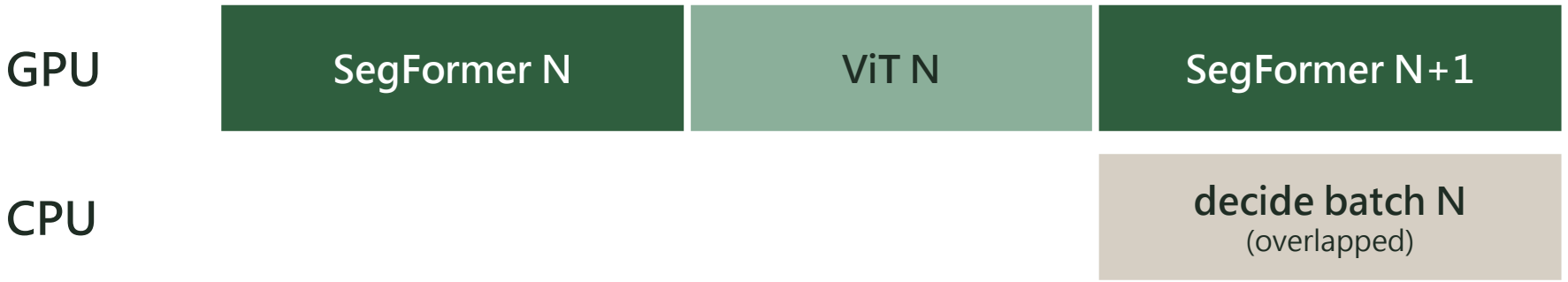


Our Journey: v8 → v9

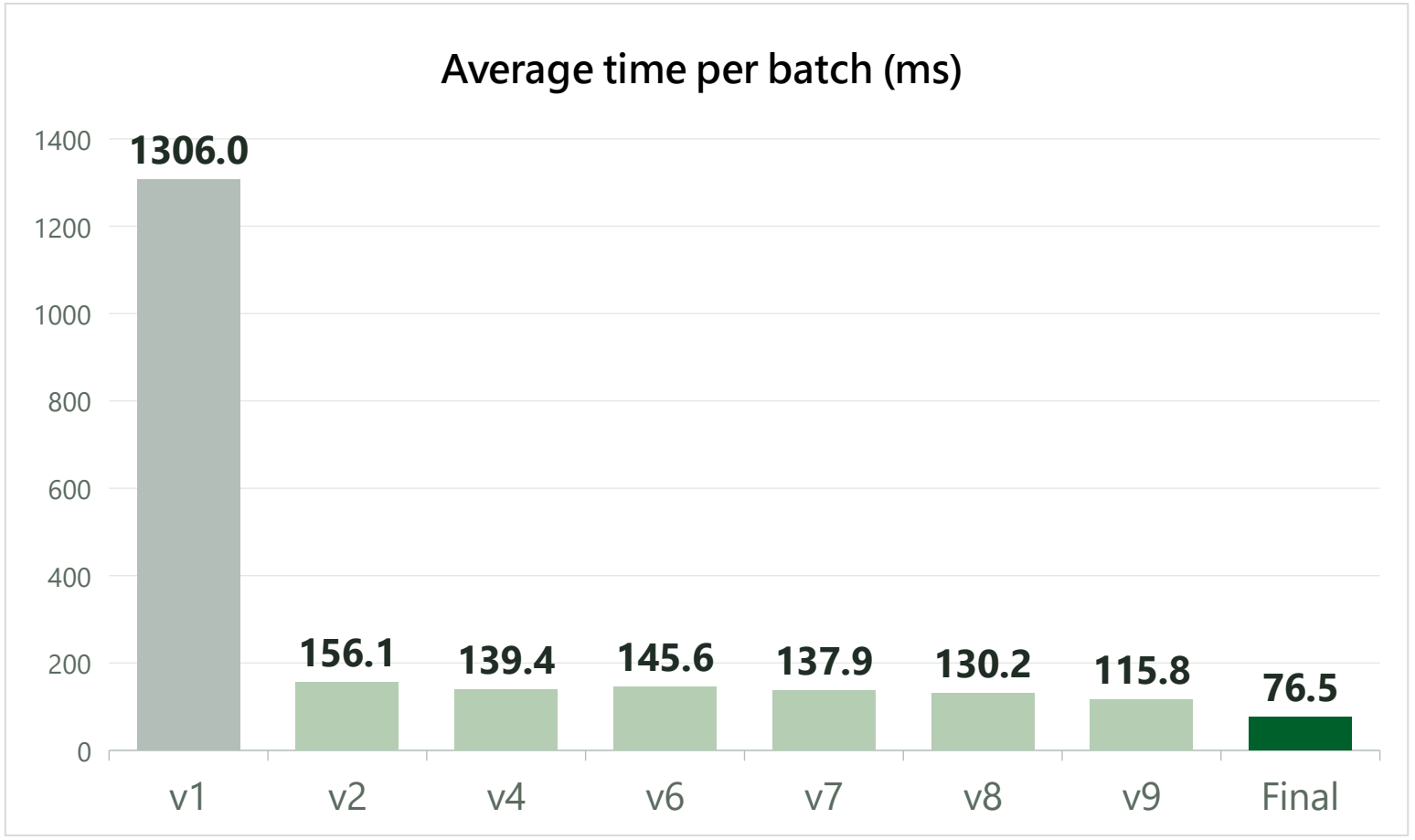
v8 — wait for ViT(N), then prepare batch N+1



v9 — send SegFormer(N+1) first, wrap up batch N afterwards



Baseline to Final: 17.1× Faster, Same Results



17.1×

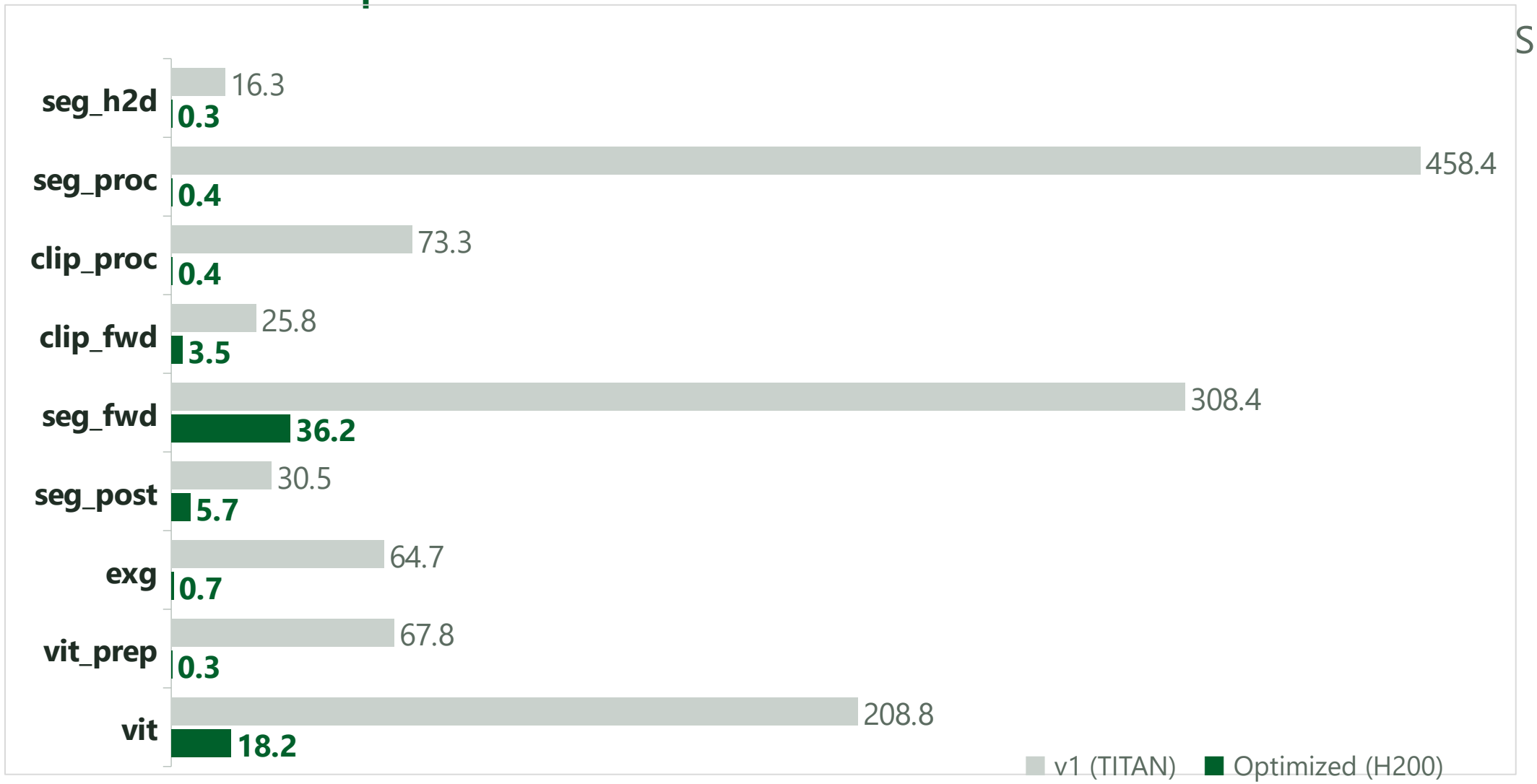
1,306 → 76.5 ms/batch

GPU active time 45.3%→ 92.2%

Identical results

Match with baseline, 1,600 images

Profiler Output: v1 vs Final



ms/batch, batch size 32, 384×384

Energy Efficiency

INPUTS		
Baseline	GPU	*Select the baseline as GPU or CPU*
Baseline GPU Type	8x V100 32GB SXM	*Select the CPU/GPU Type or GPU Node configuration*
Baseline GPU # GPUs	1	*Select the number of CPU cores or GPUs (within one node)*
Final GPU Node	8x H200 141GB SXM5	*Select the final GPU Type or GPU Node configuration*
Final # GPUs	1	*Select the number of GPUs (within one node)*
Application Speedup	17.1x	*Enter the amount of achieved speedup*

Node Replacement	17.1x

GPU NODE POWER SAVINGS			
	Baseline	**Comparison**	
	8x V100 32GB SXM	8x H200 141GB SXM5	**Power Savings**
Compute Power (W)	59,745	10,298	49,447
Networking Power (W)	2,185	880	1,305
Total Power (W)	**61,930**	**11,178**	50,752

Node Power efficiency	**5.5x**

ANNUAL ENERGY SAVINGS PER GPU NODE			
	8x V100 32GB SXM	8x H200 141GB SXM5	**Power Savings**
Compute Power (kWh/year)	523,366	90,210	433,156
Networking Power (kWh/year)	19,140	7,709	11,431
Total Power (kWh/year)	**542,506**	**97,919**	**444,587**

$/kWh	$ 0.18
Annual Cost Savings	$ 80,025.69
3-year Cost Savings	$ 240,077.07

Metric Tons of CO2	315
Gasoline Cars Driven for 1 year	68
Seedlings Trees grown for 10 years	5,211

Same 7M-images, one H200

88%less GPU time

per image, from the **17.1×** same-GPU speedup

	Base	**Optimized**
Processing time	79.4 h	**4.6 h**
Power	49.5 kW	**11.2 kW**

How AI Agents Changed Our Work

~16x

Without AI agents

1year

With AI agents

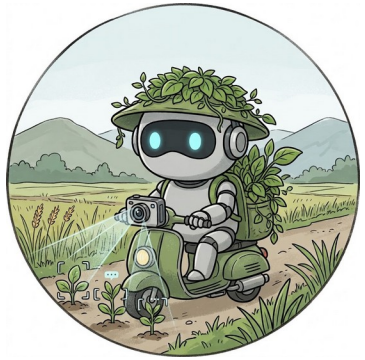
3 weeks

Before	With AI agents
CODE	
Hand-write every line	**Review and verify code**
OUR TIME	
Mostly spent coding	**More time to think and discuss deeper problems**

Acknowledgement



OpenACC More Science, Less Programming
OPEN HACKATHONS
NCHC
NARLabs 財團法人國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing
NVIDIA



Thank You

CropSight
Questions & suggestions